Predicting Poverty Vulnerability in California Using Machine Learning

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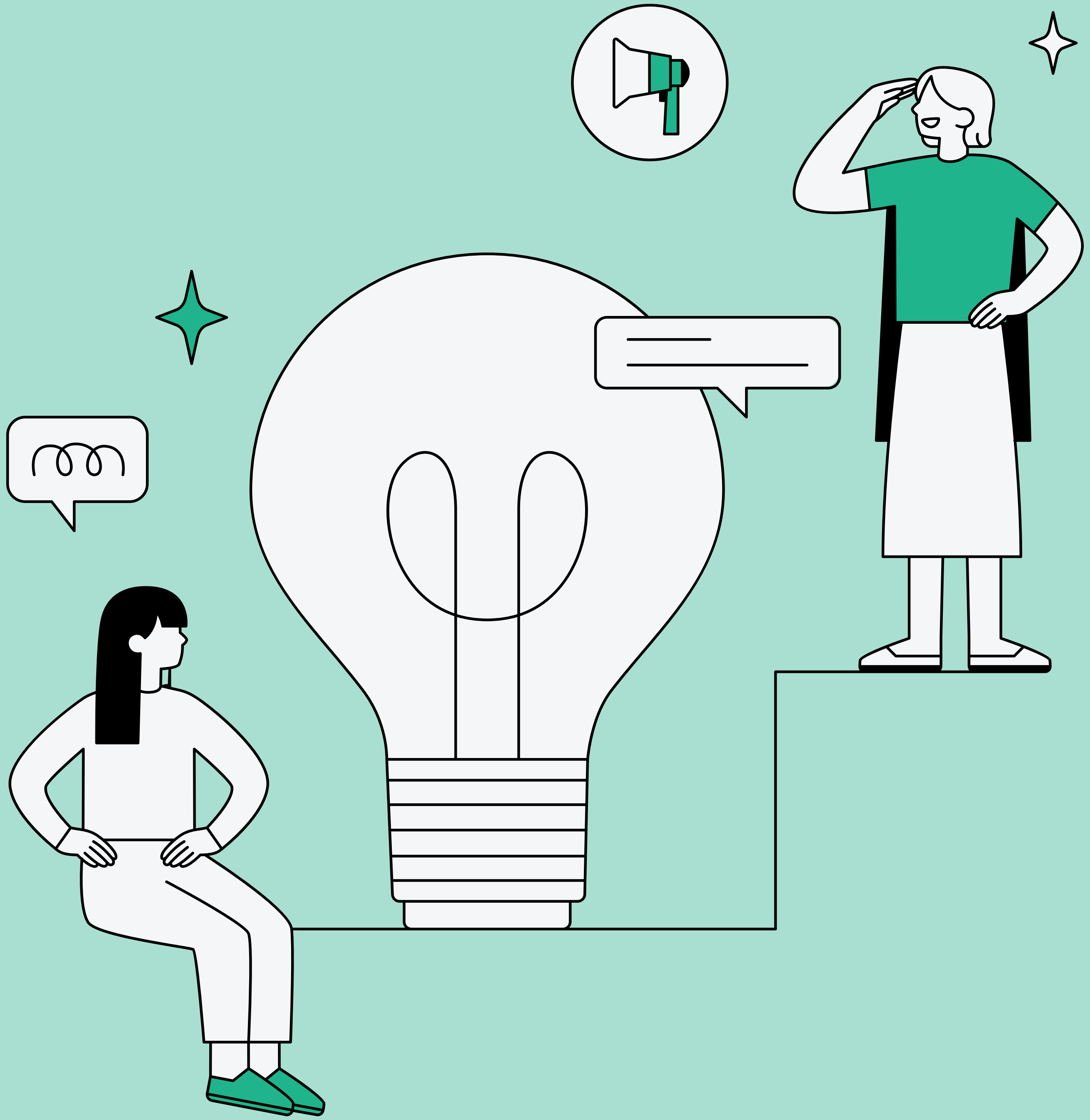
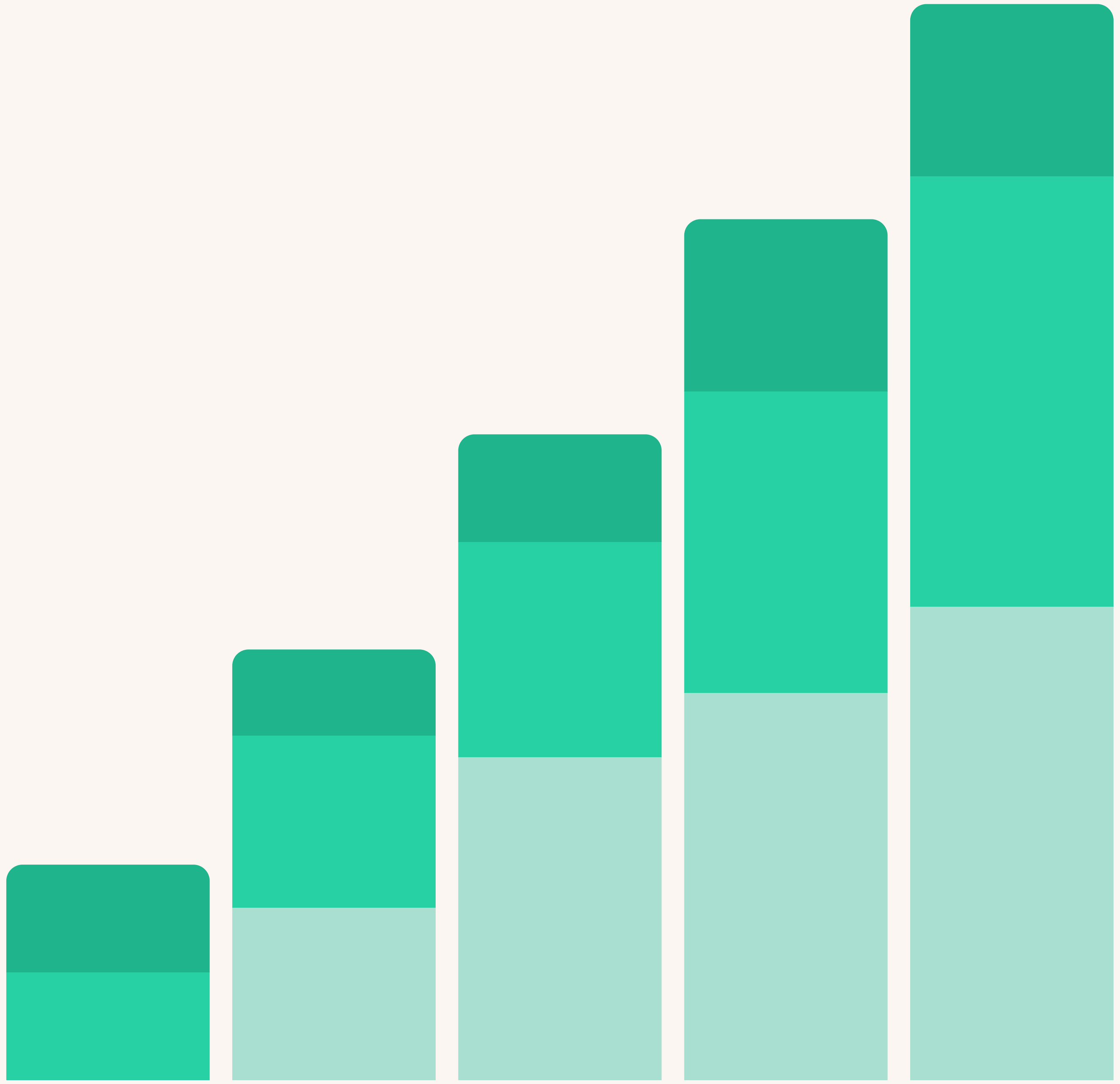
Problem & Motivation

- Poverty prediction in California
- Goal: Identify individuals at risk before they fall into poverty
- Use case: Policy + resource allocation
- Target variable: poverty_status — binary indicator derived from income-to-poverty ratio (POVPIP)
- Predictive Question: Which individuals are most likely to fall below the poverty threshold?



Data & Preparation



- Source: ACS 5-Year PUMS 2024, California
- 27 features selected across education, employment, income, demographics, and health insurance
- 25,000 rows sampled from 285-variable raw file using DuckDB — final dataset: 23,266 rows
- Cleaning: dropped FER (77% missing), employment vars filled with 0, DRIVESP/SCHL imputed with median, LANX/ENG with mode
- 80/20 stratified train/test split: 18,612 train / 4,654 test — 12.3% poverty class

Model Selection

Logistic Regression → baseline
Random Forest → nonlinear relationships
XGBoost → final model
Primary metrics: ROC-AUC and poverty class recall —
overall accuracy is misleading with 87.7% majority class

Model Selection Strategy

01.

Logistic Regression

- Interpretable baseline model
- Accuracy: 88%, ROC-AUC: 0.840, Poverty Recall: 0.17
- Struggles with class imbalance (low recall for poverty)
- Best for understanding feature impact via coefficients

02.

Random Forest

- Captures nonlinear relationships
- Improved performance over baseline — AUC: 0.853, Poverty Recall: 0.62
- Identifies feature importance effectively
- More robust than Logistic Regression

03.

XGBoost

- Boosting framework — each new tree corrects the error
- Highest ROC-AUC: 0.860, Poverty Recall: 0.83
- Handles class imbalance with scale_pos_weight = 7.13
- Best balance between predictive performance, minority class detection, and interpretability

Model Performance Comparison

Metric	Logistic Regression	Random Forest	XGBoost
Overall Accuracy	88.0%	83.6%	76.9%
ROC-AUC	0.840	0.853	**0.860 ✓**
Poverty Recall	0.17	0.62	**0.83 ✓**
Poverty Precision	0.51	0.39	0.33
Poverty F1	0.26	0.48 ✓	0.47
False Negatives	472	220	**100 ✓**
False Positives	96	543	974

✓ = best value for that metric | Overall accuracy is misleading due to 87.7% non-poverty class — focus on recall and AUC

XGBoost Tuning

Best params:
n_estimators=359
max_depth=3
learning_rate=0.059
subsample=0.847
colsample_bytree=0.727

01. RandomizedSearchCV optimized directly for poverty recall across 30 iterations, 5-fold cross-validation.

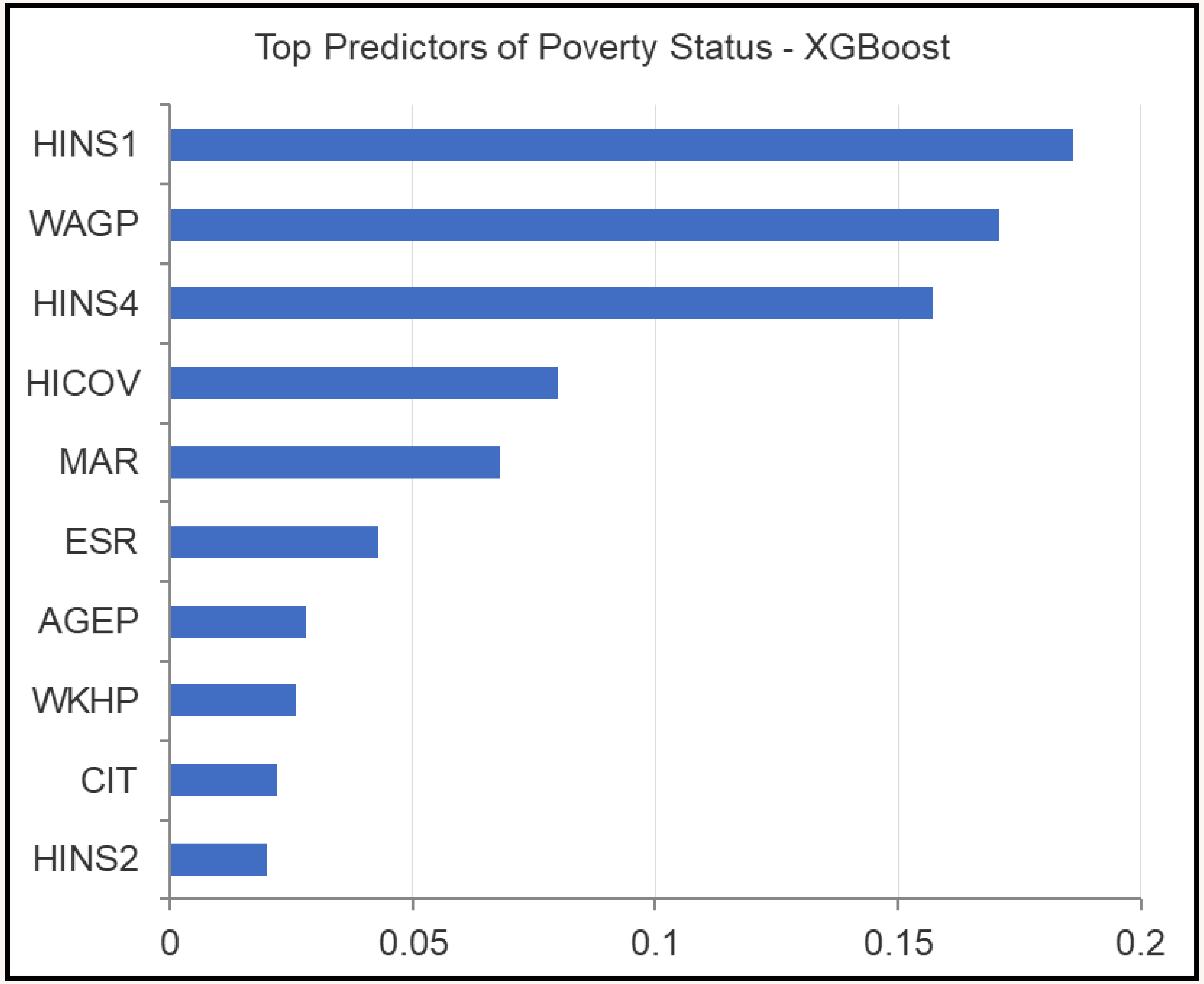
02. scale_pos_weight = 7.13 — treats each poverty case as 7x more important.

03. Shallower trees with a lower learning rate reduce overfitting on imbalanced data.

Performance Lift

Metric	Before	After
Poverty Recall	0.83	0.85
ROC-AUC	0.860	0.860
False Negatives	100	85

Final Results + SHAP



- Health insurance (HINS1, HINS4) and wage income (WAGP) are the top predictors (accounting for over 50% of XGBoost feature importance)
- Marital and employment status significantly impact vulnerability
- Age and hours worked have moderate effects
- Across all three models, the same drivers emerge consistently — confirming these are genuine patterns, not model artifacts

Streamlit Demo

California Poverty Risk Predictor

Enter demographic and employment characteristics below to estimate the probability that an individual falls below the U.S. poverty threshold. Inputs are based on American Community Survey (ACS) variables, and dropdown menus include plain-English explanations of each Census category.

Enter Individual Characteristics

Age (AGEP)

35

Education Level (SCHL)

1 = No schooling completed

Usual Hours Worked Per Week (WKHP)

40

Annual Wage Income (WAGP)

30000

Employment Status (ESR)

1 = Working currently (full-time, part-time, self-employed, gig work)

Answering the Predictive Questions

Q: Which individuals are most likely to fall below the poverty threshold?

A: Individuals with low wage income, no employer-sponsored insurance, part-time or no employment, who are unmarried, and enrolled in Medicaid are most at risk. XGBoost identifies 83% of these individuals correctly.

Q: How do education, employment, hours worked, and wages influence poverty likelihood?

A: Higher wages and full-time employment are the strongest protections. Underemployment (low WKHP even when employed) increases poverty risk independently of income. Education's effect is mainly indirect — through wages and employment type.

Q: How do age, marital status, citizenship, and insurance create differences in risk?

A: Unmarried individuals face higher risk due to lack of income pooling. Non-citizens and non-English speakers face structural barriers. Younger workers and older individuals without retirement income are most vulnerable. Medicaid enrollment signals existing vulnerability.

Conclusion & Takeaways

- XGBoost achieved the best performance (highest recall + ROC-AUC)
- Health insurance and income are the strongest drivers of poverty risk
- Employment stability and access to resources significantly impact vulnerability
- Model highlights structural factors beyond just income
- Limitations: class imbalance, cross-sectional snapshot, no regional cost-of-living adjustment
- Future work: threshold tuning, geographic variables (PUMA), multi-year ACS panel

Thank you